Restore-Sensitive Support Normal Forms for Multi-Tag Quantum Compilation

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Abstract

Support ceilings in exact quantum compilation can describe the reach of a normalization argument rather than the compiler's intrinsic optimum. We identify the semantic and objective conditions under which a zero-sum deletion argument gives a valid simultaneous normal form for an explicit multi-Tag extension of Tag-and-Restore Encoding (TARE). Each constrained frame is assigned an admissible signature alphabet before optimization. If every zero-sum deletion preserves whole-instance feasibility, does not increase the objective, and cannot activate support elsewhere, a terminating global descent produces one optimum satisfying every frame ceiling simultaneously. In the elementary binary signature groups used by the grammar, the longest zero-sum-free sequence over an alphabet that generates the group equals the group's rank. Deleting one frame coordinate changes one argument of one charged b-way Restore term, whose exact upward sensitivity is $b-1$. The descent is therefore objective-nonincreasing when $\mu \geq (b-1)t_R$, and every admitted instance has an optimum with frame support at most $s+1$ for s shared Tags. For a matched one-Tag, three-block family, an exact support-one obstruction makes the support-two ceiling intrinsic. The theorem is conditional on the stated grammar and incidence contract; it establishes neither general MultiTag sharpness nor a physical quantum advantage.

Keywords: quantum compilation; Pauli strings; support normal forms; zero-sum deletion; sparse optimization

1. Scientific question and contribution

Tag-and-Restore Encoding (TARE) maps a mutually anticommuting Pauli family to target Pauli strings and uses shared Tag operators to identify the branches [1]. Its Tag equations can have multiple solutions, and the original construction explicitly notes minimum-weight and joint optimization as possible choices. This flexibility raises a separate question: when does a local deletion certificate imply a support normal form for the complete compiler objective?

Classical zero-sum theory already studies the sequence length at which a nonempty zero-sum subsequence is forced, including variants restricted to a subset of an abelian group [2-4]. General sparse-support results are also known for integer optimization [5]. Neither body of work supplies the compiler-specific semantic step: a zero-signature deletion must remain feasible for the whole instance, and its effect on every charged term must be controlled.

This paper establishes four bounded results:

1. a whole-instance deletion contract and a global descent that yield simultaneous support ceilings;
2. an exact binary identity for the alphabet generated by the local signatures;
3. the exact Restore sensitivity $b-1$ and the resulting objective cone for the stated MultiTag grammar; and

4. a matched one-Tag, three-block family F_M in which separate upper and lower arguments show that the support-two ceiling is intrinsic.

The zero-sum invariant, its binary specialization, and general sparse optimization are used as prior mathematics. The paper does not claim a new Davenport-type invariant, general MultiTag sharpness, production-compiler transfer, or a hardware-resource advantage.

2. Fixed alphabets and zero-sum-free sequences

Let H be a finite abelian group written additively, and let $A \subseteq H$ be a finite alphabet fixed independently of the optimum to be bounded. A subsequence selects an arbitrary set of positions and need not be contiguous. A sequence over A is **zero-sum-free** if no nonempty subsequence has sum zero. Define

$$\mathrm{zsf}(H; A) = \max(\{0\} \cup \{|W| : W \text{ is zero-sum-free over } A\}).$$

The explicit zero handles degenerate alphabets. If $D(H)$ denotes the least length forcing a nonempty zero-sum subsequence over the full finite group, then

$$\mathrm{zsf}(H; H \setminus \{0\}) = D(H) - 1.$$

For restricted A, $\mathrm{zsf}(H; A) + 1$ is the corresponding forcing length. This is donor mathematics, not a contribution of the present paper.

3. Simultaneous deletion theorem

Fix an optimization instance with a nonempty finite feasible set, so an exact optimum exists. For every constrained generator R, let A_R be the set of signatures realizable by any admissible local state of that instance, fixed before optimization. Let every active coordinate q of R carry $v_q \in A_R \subseteq H_R$.

Assume the following.

1. **Nonzero total.** Every feasible constrained generator satisfies $\sum_q v_q \neq 0$.
2. **Whole-instance deletion soundness.** Deleting any nonempty zero-sum subsequence of coordinates of R preserves every constraint of the full instance, not only constraints local to R.
3. **Objective dominance.** The same deletion does not increase the objective.
4. **Support monotonicity.** Deleting coordinates of one generator does not activate coordinates of another generator or otherwise increase any generator's support.

Theorem 1 (simultaneous alphabet ceiling). Every admitted instance has an exact optimum $x^\star$ such that, simultaneously for every constrained generator R,

$$\mathrm{support}_R(x^\star) \leq \mathrm{zsf}(H_R; A_R).$$

Proof. Start from any optimum. While some generator contains a nonempty zero-sum subsequence, delete that subsequence. Its nonzero total excludes deletion of the whole active sequence. Assumptions 2 and 3 preserve feasibility and optimality. Assumption 4 makes total support over all constrained generators decrease strictly and prevents any individual support from increasing. Total support is a nonnegative integer, so the process terminates. At termination every active signature sequence is zero-sum-free over its fixed alphabet and hence has length at most the corresponding zsf value. The same terminal optimum therefore satisfies all generator bounds. □

The global descent is essential: reducing one generator only once would not establish the existential-optimum/universal-generator quantifier when generators share constraints.

4. Exact binary specialization

The restricted alphabet does not improve the rank ceiling once the binary signature group is defined to be the span of that alphabet.

Theorem 2 (binary generated-span identity). Let $A \subseteq \mathbb{F}_2^d$, and let $H = \langle A \rangle$. Then

$$\mathrm{zsf}(H; A) = \mathrm{rank}(H).$$

Proof. A sequence over A is zero-sum-free exactly when its listed vectors are linearly independent: any binary linear dependence selects a nonempty set of positions whose bitwise exclusive-or (XOR) is zero, and the converse is immediate. Thus every zero-sum-free sequence has length at most $\mathrm{rank}(H)$. Because A spans H, it contains a basis of H; listing that basis gives a zero-sum-free sequence of length $\mathrm{rank}(H)$. □

This equality is special to the elementary binary setting. It must not be generalized to arbitrary finite abelian groups. For example, in $\mathbb{Z}_n$ with alphabet $\{1\}$, the sequence of $n-1$ ones is zero-sum-free although the group has rank one.

The equality is still only a certificate ceiling. It becomes an intrinsic compiler value only when a separate compiler witness rules out every smaller-support optimum.

5. Phase-quotiented MultiTag-TARE parity grammar

We now define the formal compiler grammar to which the abstract deletion theorem is applied. An n-qubit Pauli is represented up to phase by $P = (x, z) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$. Multiplication is componentwise XOR, support is

$$w(P) = |\{q : (x_q, z_q) \neq (0, 0)\}|,$$

and the binary symplectic product is

$$\langle (x, z), (x', z') \rangle = x \cdot z' + z \cdot x' \pmod 2.$$

Fix $b \geq 2$ ordered blocks, a finite set of frame roles u, and $s \geq 0$ shared Tag Paulis $S_1, \ldots, S_s$. Block ℓ contains a frame $R_{\ell u}$ for each role and a designated partner $R'_{\ell u}$. The parity constraints involving that frame are

$$\langle R_{\ell u}, R'_{\ell u} \rangle = 1, \qquad \langle S_j, R_{\ell u} \rangle = \lambda_{j,\ell u} \quad (j = 1, \ldots, s),$$

where the required label bits $\lambda_{j,\ell u}$ are part of the instance. These parity equations are the only feasibility constraints in which the letters of $R_{\ell u}$ participate. Targets, central-role choices, and any other variables are held fixed during a frame deletion, although they may be optimized before the deletion step is selected.

For an active coordinate q of a frame $R = R_{\ell u}$, define

$$v_q = (\langle R_q, R'_q \rangle, \langle S_{1q}, R_q \rangle, \ldots, \langle S_{sq}, R_q \rangle) \in \mathbb{F}_2^{s+1},$$

where $\langle\cdot,\cdot\rangle$ is the local binary symplectic product. The XOR of the first components is

$$\sum_q \langle R_q, R'_q\rangle = \langle R, R'\rangle = 1,$$

because the global symplectic product is the sum of its local products and R anticommutes with R'. The remaining components sum to the prescribed Tag labels. The total signature is therefore nonzero.

For each frame R, define A_R from all local signatures allowed by the instance grammar, not from a selected optimum, and set $H_R = \langle A_R\rangle$.

5.1 Deletion semantics

For $Q \subseteq \{1,\ldots,n\}$, deletion of Q from R replaces R_q by I for every $q \in Q$ and changes no other variable. The change in the partner and Tag parity vector is exactly

$$\bigoplus_{q\in Q} v_q.$$

Consequently a zero-signature deletion preserves every parity equation of the grammar. Because no other feasibility constraint contains the deleted letters, it preserves whole-instance feasibility. It only turns active letters into identities, so it cannot increase any frame support.

5.2 Restore incidence and objective

Each frame role u has fixed target Paulis $T_{1u},\ldots,T_{bu}$. At coordinate q, its Restore term takes the ordered residual letters

$$a_{\ell uq} = (T_{\ell u}R_{\ell u})_q, \qquad \ell = 1,\ldots,b.$$

For each pair $(R_{\ell u}, q)$, this is the only Restore term containing the letter $(R_{\ell u})_q$. Deleting that letter replaces exactly the ℓ-th argument of this one term, from $(T_{\ell u}R_{\ell u})_q$ to $(T_{\ell u})_q$.

The objective is

$$C = C_0 + C_{\mathrm{Tag}}(S_1,\ldots,S_s) + \sum_{\ell,u,q} \mu_{\ell uq}\mathbf{1}[(R_{\ell u})_q \neq I] + t_R \sum_{u,q} F_b(a_{1uq},\ldots,a_{buq}),$$

where C_0 is independent of the frame letters, $C_{\mathrm{Tag}} \geq 0$ is unchanged by a frame deletion, $\mu_{\ell uq} \geq \mu$, and $t_R \geq 0$. The coefficients $\mu_{\ell uq}$ may depend on a central-role or target-order choice, but that choice is held fixed while the deletion is evaluated and the lower bound μ is uniform over all allowed choices.

This one-occurrence incidence is part of the formal grammar. A compiler in which one frame deletion changes several Restore terms requires a different coefficient and is outside the theorem below.

6. Exact Restore sensitivity

For a local Pauli alphabet containing the identity and at least two distinct nonidentity letters, define

$$F_b(a_1,\dots,a_b)=\begin{cases}1, & a_1=\cdots=a_b\neq I,\\ |\{i: a_i\neq I\}|, & \text{otherwise.}\end{cases}$$

Lemma 3 (one-argument sensitivity). Replacing one argument of F_b can increase its value by at most $b-1$, and this bound is attained.

Proof. Away from the all-equal nonidentity state, changing one argument raises ordinary nonidentity Hamming weight by at most one; entering the exceptional state lowers the value. Leaving an all-equal nonidentity state by replacing one letter with a different nonidentity letter changes the value from 1 to b, an increase of $b-1$. No case is larger. □

7. Restore-sensitive MultiTag normal form

Deleting k coordinates refunds at least $k\mu$. By the incidence contract and Lemma 3, the Restore increase is at most $k(b-1)t_R$. This bound is also valid when several one-argument replacements affect the same local functional: order the replacements and telescope the one-argument bound. The deletion is therefore objective-nonincreasing when

$$\mu \geq (b-1)t_R.$$

Theorem 4 (MultiTag support normal form). Under the stated semantic and incidence contracts, if $\mu \geq (b-1)t_R$, every admitted instance has an exact optimum satisfying

$$\mathrm{support}(R) \leq \mathrm{zsf}(H_R; A_R) = \mathrm{rank}(H_R) \leq s+1$$

simultaneously for every constrained frame R.

Proof. Section 5.1 proves whole-instance feasibility and support monotonicity for a zero-signature deletion. The cost calculation above establishes objective dominance. Theorem 1 gives the simultaneous zsf ceiling. Since $H_R = \langle A_R\rangle$ is an elementary binary subgroup of $\mathbb{F}_2^{s+1}$, Theorem 2 gives equality with its rank, which is at most $s+1$. □

Outside the stated cone, this proof is unavailable. That fact does not establish that larger support is necessary.

8. Matched support-two control

The matched family F_M is the one-Tag, three-block specialization with

$$b=3, \qquad s=1, \qquad \mu=2, \qquad t_R=1.$$

Its objective lies on the cone boundary. The signature group has rank two, so Theorem 4 gives an all-size support-two upper bound. The matching lower direction is witnessed at $n=2$. Using the binary Pauli keys (x,z) of Section 5, the target pairs and two feasible frame states are shown below. Each integer in a key is the decimal encoding of a two-bit mask: for example, $2 = 10_2$ selects the second qubit and $3 = 11_2$ selects both qubits. Thus $(0,2)$ and $(3,0)$ denote two-qubit Paulis, not individual binary coordinates.

Block	Target pair $(T_{\ell 0}, T_{\ell 1})$	Feasible support-two frame pair	A minimizing support-at-most-one frame pair
A	$((0,1),(0,1))$	$((0,1),(1,1))$	$((0,1),(1,0))$
B	$((0,1),(0,1))$	$((0,1),(1,1))$	$((0,1),(1,0))$
C	$((0,2),(2,0))$	$((0,2),(3,0))$	$((0,1),(1,1))$

Both states use $S = (0,1)$. For the support-two state, the central choices are $(0,0,1)$ and the target orders are unchanged. Its weighted-frame, Tag, constant, and Restore contributions are $20, 2, -18, 1$, giving total cost 5. The displayed support-at-most-one state has contributions $18, 2, -18, 4$, giving cost 6.

To establish the lower bound, the checker enumerates the complete support-at-most-one family: all $7^6 = 117{,}649$ six-frame tuples, including all $12^3 = 1{,}728$ anticommuting tuples, all 16 Tags, all eight central choices, and all four allowed relative target orders. The exact minimum is 6. Because the feasible support-two state costs 5, no support-at-most-one state can be optimal for this instance. Hence support one cannot be a uniform optimum bound, and

$$\kappa(F_M; C_M) = 2.$$

Here κ is the least uniform support bound attained by exact optima under this fixed objective. The lower witness makes the certificate ceiling intrinsic for F_M; no analogous lower theorem is claimed for general MultiTag-TARE.

9. Discussion and relation to prior work

The closest zero-sum donor is the subset-conditioned Davenport literature [2], with broader generalized and weighted variants providing context [3,4]. In the finite binary grammar considered here, however, the alphabet-specific sequence ceiling collapses exactly to the rank of its generated span. The novelty cannot therefore rest on a smaller restricted-alphabet bound. It rests on the compiler transfer: the admissible alphabet is fixed before optimization, deletion is sound for the whole instance, the descent closes the simultaneous quantifier, and the Restore incidence exposes the exact objective condition.

Sparse integer optimization provides general support bounds for optimal solutions [5], while binary symplectic representations of Pauli operators are standard [6,7]. The present result is not a generic sparsity theorem or a new stabilizer formalism. It identifies when a particular local deletion operation preserves the semantics and objective of the stated compiler grammar.

The matched family also clarifies the difference between a certificate ceiling and intrinsic support. The rank argument alone supplies only an upper bound. Equality with intrinsic support follows here because a separate exact instance excludes support one. A compiler with transformations outside the deletion language may have a strictly smaller intrinsic value, a possibility treated explicitly in the companion analysis of certifiable support budgets.

10. Reproducibility

The accompanying source includes a deterministic checker derived from the theorem statements. It enumerates all alphabets in $\mathbb{F}_2^d$ for $d = 1, 2, 3$, checks the generated-span equality, tests the cyclic counterexample to a general rank reading, verifies zero-sum deletion and global descent on finite fixtures, and exhausts Restore sensitivity for $b = 2, \ldots, 7$. The source also includes the exact two-site support-one obstruction summarized in Section 8.

These computations corroborate finite case analyses and transcription. They do not replace the all-size proofs, establish implementation transfer, or constitute external replication.

An earlier finite production-realization study of the matched family rejected its proposed certificate. The declared optimizer values were corroborated, but the study did not establish a complete production move registry; a source-bound follow-up again left production transfer unestablished. These are adverse transfer results, not failures of the formal support theorem. Their exact machine terminals and immutable evidence remain in the accompanying provenance ledger.

A separate, prospectively ordered production-transfer attempt used pinned PyZX 0.10.5. It executed 74 of 4,681 ordered two-qubit circuit words before failing closed on the word consisting of three consecutive Hadamard gates on qubit 0. After a semantics-preserving prefix, the callable guard accepted a second boundary-pivot simplification, but that transition changed the dense linear map even up to a nonzero scalar. The scheduled production reduction preserved the map on the same word. This adverse result refutes only the proposed freely reorderable language of twelve macros under their callable guards. It does not refute the scheduled production reduction, establish complete PyZX move coverage, realize a certificate gap, or authorize a complete-domain null. The round is consumed as an indeterminate completeness study and is not retuned or relabelled; its exact machine-readable terminal is retained in the provenance ledger.

11. Limitations

The normal-form result is conditional on whole-instance soundness, objective dominance, and support monotonicity. Its MultiTag specialization further uses the one-deletion/one-Restore-argument incidence stated in Section 5.2. The binary rank identity is specific to the generated binary signature groups and does not extend to arbitrary finite abelian groups. The proof is also silent outside $\mu \geq (b-1)t_R$: it implies no larger-support necessity there, and sharpness for general MultiTag instances remains open.

These scope restrictions also limit the result's interpretation. Structural support is not T count, depth, runtime, qubit count, error rate, or quantum advantage. The adverse PyZX transfer record shows that callable macro guards omit load-bearing scheduler context, so the formal families have not been shown to reproduce a complete production move language. Finally, the author-side checks reported here do not constitute independent proof certification, novelty authority, peer review, or external replication.

12. Conclusion

A zero-sum deletion ceiling becomes a compiler normal form only after its semantic and objective obligations are proved. For the stated MultiTag grammar, global anticommutation supplies a nonzero signature total, the coordinate-indexed Restore term has exact sensitivity $b-1$, and a global descent gives one optimum satisfying all frame ceilings. Because each binary signature group is defined as the span of its alphabet, the zero-sum-free ceiling equals rank rather than improving it. The matched family F_M supplies a bounded case in which a separate lower witness makes the support-two certificate intrinsic.

Data and code availability

The submission source contains the deterministic finite checker, its expected output, the control records used for the exact two-site obstruction, and the immutable adverse PyZX transfer record summarized above. No external dataset is used. The records are provenance aids rather than theorem authority; the displayed proofs and exact obstruction carry the stated all-size and lower-bound claims. The exact source archive is available to editors and referees with the manuscript and will accompany the public manuscript record.

Author contributions

Sze Chun Yiu is the sole author and is responsible for the scientific claims and final manuscript. Generative AI tools assisted literature discovery, drafting, language editing, adversarial review, and submission-package preparation.

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